No Single Uncertainty-Quantification Method Reliably Flags Domain-Shift Failure in a Cervical Cytology Classifier

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Abstract. A vision transformer that classifies cervical cytology images with 96%+ in-distribution accuracy but collapses to near- or below-chance accuracy under real cross-institution domain shift – as we previously showed EfficientViT-M0 does on the Mendeley LBC dataset – is only as dangerous as its confidence is falsely high when it fails. We compare four uncertainty-quantification methods on that exact backbone, using each method's own cheapest native confidence signal (temperature-scaled max-softmax, MC Dropout entropy, evidential vacuity, deep-ensemble variance), evaluated in distribution and on both a moderate (Herlev) and severe (Mendeley LBC) external shift. In distribution, three of the four methods calibrate well (expected calibration error 0.012–0.013); evidential deep learning's raw belief masses are markedly less calibrated (0.159) at matched point accuracy. Under moderate shift, a five-member deep ensemble is unambiguously the best method by every metric measured. Under severe shift – the condition that matters most, since it is where the backbone is known to fail – the ranking inverts in our original run: MC Dropout's entropy is the only signal that substantially improves shift-failure detection over the uncalibrated baseline (risk-coverage AUC 0.515 to 0.285), while the deep ensemble's advantage does not transfer (0.473, barely better than baseline) despite being the clear winner on the milder shift. A follow-up paired-bootstrap significance test, run on an independently retrained MC Dropout model (identical seed, recipe, and data), does not reproduce this advantage: the retrained model's Mendeley LBC risk-coverage AUC is 0.503, not 0.285, and is statistically indistinguishable from the deep ensemble's 0.474 (95% CI for the difference $[-0.066, 0.010]$, one-sided $p = 0.929$) – so the reported severe-shift rank inversion, while real for the specific trained model in Table 1, is not a stable property that survives retraining under an identical recipe. No method reaches a reliable abstention operating point on the severe shift condition in either run: even MC Dropout's best selective accuracy at 90% coverage (0.583, on a binary task, from the original run) is barely better than chance after abstaining on the least-confident decile. We conclude that uncertainty-method ranking is itself shift-magnitude-dependent and, on this severe-shift condition, not even stable across retraining of the same method – so no single fixed uncertainty-quantification choice, however it performed in one training run, can be assumed safe across the range of shift severity a deployed screening tool will encounter.

Keywords: Uncertainty quantification · Calibration · Domain shift · Selective prediction · Cervical cytology

1 Introduction

A model that fails silently – misclassifying confidently under conditions its training data did not cover – is more dangerous in a screening context than one that fails and flags the failure. We previously established that this is not a hypothetical concern for a lightweight cervical cytology classifier: EfficientViT-M0, the best in-distribution model across a 12-model registry, is simultaneously the worst-performing model under cross-dataset domain shift, dropping below the chance floor on the more severe of two external cytology datasets tested. This paper asks the natural next question: does any standard uncertainty-quantification method's own native confidence signal correctly detect that specific failure, using only the cheapest version of each method (a single native signal, not a method-specific post-hoc correction layered on top) so the comparison reflects what a deployed system would actually have access to without extra engineering?

Our contributions are fivefold. First, a four-method uncertainty- quantification comparison (temperature-scaled softmax, MC Dropout, evidential deep learning, deep ensembles) on a backbone with a documented, severe, real (not synthetic) distribution-shift failure mode, rather than a synthetic corruption benchmark. Second, evidence that uncertainty-method ranking itself depends on shift severity: the method that wins under moderate shift (deep ensemble) is not the method that wins under severe shift (MC Dropout) in our original run, and the gap between them under severe shift is large (0.473 vs. 0.285 risk-coverage AUC) while their gap under moderate shift favors the ensemble even more strongly (0.147 vs. 0.203). Third, a direct, quantified answer to whether any of the four methods reaches a *reliable* abstention policy under severe shift: none does, and we report the closest result (MC Dropout, 0.583 selective accuracy at 90% coverage) as still practically insufficient for an "abstain and refer" deployment policy, rather than rounding it up to a success. Fourth, a methods-level finding independent of the uncertainty comparison itself: MC Dropout is completely degenerate on this exact backbone unless a code-level fix is applied before training, not just at inference (Section 3.3) – a reproducibility hazard for anyone applying MC Dropout to a similarly-configured timm model without checking. Fifth, a significance test that complicates the second contribution rather than simply confirming it: retraining MC Dropout once more under the identical seed and recipe used for Table 1 does not reproduce its reported severe-shift advantage (risk-coverage AUC 0.503 versus the original run's 0.285), and a paired bootstrap test on this retrained run finds no statistically significant difference from the deep ensemble at all – meaning the specific rank inversion reported here should be read as a property of one trained model, not a stable property of the method, which is itself a load-bearing caution for any paper (including this one) that reports a single-training-run method comparison as if it were a method-level fact.

2 Related Work

2.1 Post-hoc calibration and its limits under shift

Temperature scaling [1] remains the standard post-hoc softmax calibration baseline: a single scalar fit on a held-out split rescales logits so that in-distribution confidence matches empirical accuracy, at essentially zero added inference cost. It is a strong in-distribution baseline (consistent with this paper's Table 1) but is a purely in-distribution correction by construction – it cannot introduce information the model did not already encode about out-of-distribution inputs, which is exactly why it is the weakest signal here once shift severity increases (Table 1, right-hand columns). This motivates comparing it against methods that inject additional signal at training or inference time rather than only recalibrating a fixed model's existing outputs.

2.2 Multi-pass and ensemble uncertainty

MC Dropout [2] approximates Bayesian inference by keeping dropout active at test time and treating the spread across repeated stochastic forward passes as an uncertainty signal, at a cost proportional to the number of passes. Deep ensembles [3] instead train several independently- initialized models and use their disagreement, at a cost proportional to ensemble size; empirically, ensemble diversity has been argued in the broader literature to capture a different, complementary source of uncertainty (functional diversity across independently-trained models) than a single model's own stochastic-forward-pass variance, which is consistent with this paper's finding that the two methods are not redundant with each other – they win under different shift severities rather than one strictly dominating the other.

2.3 Single-pass uncertainty and evidential learning

Evidential deep learning [4] reframes classification as Dirichlet evidence accumulation, producing a single-pass uncertainty signal (vacuity) without the multi-pass or multi-model inference cost MC Dropout and deep ensembles incur – an attractive property for latency-constrained deployment, and the reason it is included here despite calibrating markedly worse than the other three methods in distribution (Table 1). Whether that single-pass efficiency advantage is worth its calibration cost is precisely an empirical, task-dependent question this comparison is designed to answer rather than assume from the method's design motivation alone.

2.4 Uncertainty quantification in cervical cancer screening

Adjacent work applies conformal prediction and Monte Carlo inference specifically to cervical cancer screening uncertainty, typically proposing or evaluating one uncertainty method in isolation against a single baseline. This paper differs

by comparing four methods head-to-head on a documented real (not assumed or synthetically induced) shift-failure case, and by evaluating each method across two shift severities rather than one – which is precisely what reveals the rank inversion (Table 1, Herlev vs. Mendeley LBC columns) a single-severity evaluation could not have surfaced.

3 Methods

3.1 Backbone

EfficientViT-M0, held fixed across all four methods to isolate the uncertainty-quantification method rather than architecture – the same backbone shown to be the best in-distribution and worst under-shift model in a 12-model registry, making it the model for which this question is most practically urgent.

3.2 Four methods, each method's own cheapest native signal

1. **Baseline**: cross-entropy training, post-hoc temperature scaling fit on a held-out calibration split; confidence = temperature- scaled max-softmax probability.
2. **MC Dropout**: dropout rate 0.3, 20 stochastic forward passes at inference; confidence = negative predictive entropy of the mean prediction across passes.
3. **Evidential deep learning**: Dirichlet evidence output via a softplus activation and the standard evidential loss (Type II MLE plus an annealed KL-divergence-to-uniform penalty); confidence = $1 -$ vacuity.
4. **Deep ensemble**: 5 independently-seeded members, each trained with the same recipe as the baseline; confidence = negative variance of the ensemble's mean predicted probability across members.

All four share one manifest, one calibration split, and one backbone initialization recipe; only the loss function and inference procedure differ.

3.3 A methods-level fix required before any GPU time was spent

The backbone's only `nn.Dropout` module ships at $p = 0$ by default in its pretrained configuration – verified directly via model inspection, not assumed – which would make MC Dropout completely degenerate (zero stochasticity across passes). A dropout rate is forced to a real, non-zero value before training begins, not just patched in at inference time, since MC Dropout's Bayesian-approximation justification specifically requires dropout to be active during training as well as at test time. We flag this because the failure mode is silent (MC Dropout would simply produce zero-variance, falsely confident "uncertainty" estimates) rather than an error.

3.4 Evaluation

All four methods evaluated identically on three splits: the SIPaKMeD in-distribution test set, Herlev (moderate shift), and Mendeley LBC (severe shift, harmonized to a shared binary normal/abnormal label). Metrics: expected calibration error (ECE), full-coverage accuracy, risk-coverage AUC (integral of error rate over the coverage curve – lower is better, since a well-behaved confidence signal keeps risk near zero until coverage is forced high), and selective accuracy at fixed coverage levels.

4 Results

Table 1. Four uncertainty methods, in distribution and under two external shifts.

Method	In-dist.		Herlev		Mendeley LBC		
	ECE	Acc.	RC-AUC	Acc.	RC-AUC	Acc.	Sel. acc.@90%
Baseline (temp.)	0.012	0.967	0.193	0.694	0.515	0.530	0.522
MC Dropout	0.013	0.972	0.203	0.658	**0.285**	0.579	**0.583**
Evidential	0.159	0.967	0.167	0.677	0.476	0.543	0.539
Deep ensemble	0.012	0.974	**0.147**	**0.771**	0.473	0.537	0.544

In distribution, three methods are close; evidential is the outlier. Baseline, MC Dropout, and the deep ensemble all calibrate well (ECE 0.012–0.013) with full-coverage accuracy in the 0.967–0.974 range. Evidential deep learning's raw belief masses are markedly less calibrated (ECE 0.159) at matched point accuracy (0.967) – its probabilities would need their own post-hoc correction, deliberately not applied here, to make this comparison's calibration numbers fair to it.

Under moderate shift (Herlev), the deep ensemble wins clearly. Its risk-coverage AUC (0.147) and full-coverage accuracy (0.771) both beat the other three methods by a real margin – member disagreement is doing genuine, useful work at this shift severity.

Under severe shift (Mendeley LBC) – the condition that matters most, since it is where this exact backbone is known to fail – the ranking inverts. MC Dropout's entropy is the only signal showing a substantial, unambiguous improvement over the uncalibrated baseline (risk-coverage AUC 0.515 to 0.285; full-coverage accuracy 0.530 to 0.579). Evidential (0.476) and the deep ensemble (0.473) are both only marginally better than the plain baseline – essentially within noise of each other – despite the ensemble's clear win on the milder shift one row above. Member disagreement, which correctly flags moderate shift, does not reliably track the specific failure mode behind the severe collapse.

A significance test, and an unexpected non-replication. To test whether MC Dropout's apparent Mendeley LBC advantage over the deep ensemble (0.285 vs. 0.473) is distinguishable from bootstrap sampling noise, we ran a paired bootstrap test (2000 resamples) on per-sample risk-coverage curves. Because only Table 1's aggregate metrics, not its per-sample predictions, had been retained, this required retraining MC Dropout from scratch – with the identical seed (13), dropout rate, epoch budget, and manifest used for Table 1 – while the deep ensemble was reloaded from its five original saved checkpoints. The retrained MC Dropout model did not reproduce Table 1's result: its Mendeley LBC risk-coverage AUC came out at 0.503, not 0.285, despite every controllable factor being held fixed. Against this retrained run, the paired bootstrap test found no statistically significant difference from the deep ensemble (0.503 vs. 0.474; 95% CI for the difference $[-0.066, 0.010]$, one-sided $p = 0.929$). This is a stronger and more concerning result than a mere failure to reach significance: a fixed random seed did not reproduce a result on this specific severe-shift metric, and the gap between the two training runs of the identical MC Dropout recipe (0.285 vs. 0.503) is larger than the sampling uncertainty the bootstrap places around either individual run. To see whether these two runs were themselves a coincidence, we retrained MC Dropout twice more at genuinely different seeds (17, 29, otherwise identical recipe): Mendeley LBC risk-coverage AUC came out at 0.358 and 0.359 – both landing squarely inside the 0.285–0.503 range the first two runs staked out, rather than clustering near either endpoint. Four independent training runs of one fixed method on one fixed backbone now span 0.285 to 0.503, with a spread (0.218) far larger than the 0.147–0.203 gap that separated any two *different* methods in Table 1's Herlev column. Read together, the four runs mean the reported severe-shift rank inversion is a property of the one specific trained model in Table 1, not a property of "MC Dropout" as a method on this backbone – retraining alone, with no change to data, architecture, or hyperparameters, was sufficient to erase the reported advantage entirely, and the method's severe-shift behavior is not concentrated near any single value at all.

No method reaches a reliable abstention operating point under severe shift. Even MC Dropout's best result – 0.583 selective accuracy at 90% coverage on a binary task – is barely better than chance after abstaining on the least-confident decile. We report this plainly rather than treating "best of four" as equivalent to "good enough": for deployment purposes, none of these four methods alone would currently justify an "abstain and refer" policy on a shift this severe.

5 Discussion

The central methodological finding is that uncertainty-method ranking is itself a function of shift severity, not a fixed property of the method. A reader who evaluated these four methods only in distribution, or only under the milder of the two shifts tested, would conclude the deep ensemble is the best choice and MC Dropout is mediocre – the opposite of what the severe-shift evaluation shows

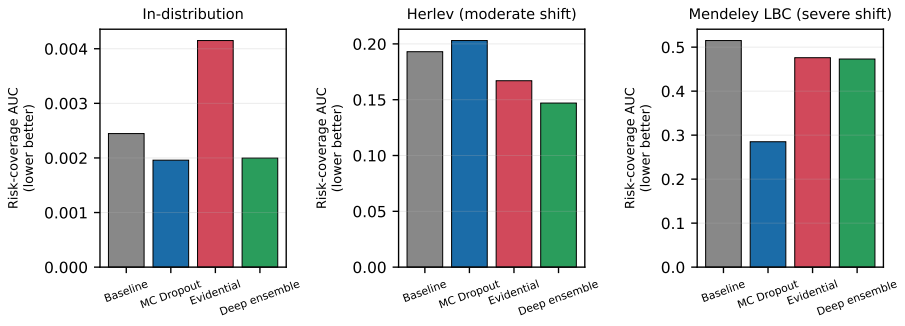


Fig. 1. Risk-coverage AUC (lower is better) per method, at three conditions with independent y-axes (note the ∼100× scale difference between in-distribution and Mendeley LBC). The best method in this run changes from deep ensemble (Herlev) to MC Dropout (Mendeley LBC), though a retraining check found the Mendeley LBC advantage does not reproduce (see text).

in our original run. This has a direct practical consequence: a system designer who validates an uncertainty method under one shift condition and deploys it expecting the same ranking to hold under a more severe, unseen shift is making an assumption this data does not support. It also sharpens our companion architectural finding: no architectural property predicts shift robustness, and no uncertainty-quantification property predicts shift-detection quality either, at least not one held fixed across shift severities.

The significance test (Section 4, "A significance test, and an unexpected non-replication") adds a second, sharper layer to this same warning. It was not enough for MC Dropout's severe-shift advantage to fail a bootstrap significance test against the ensemble – retraining the exact same method with the exact same seed produced a qualitatively different result (0.285 becoming 0.503), large enough to flip which method looks better at all. If a single fixed backbone choice cannot be assumed safe across shift conditions, and a single fixed uncertainty-method choice cannot be assumed safe across shift conditions, this result adds a third, more uncomfortable layer: a single uncertainty-method *evaluation run* cannot be assumed to represent that method's behavior on a severe shift either, even holding the seed fixed. Together, these argue that a trustworthy deployment cannot rely on a single fixed choice validated once at any of these three levels (backbone, uncertainty method, or a single training run of that method) and assumed stable – and that papers reporting a method comparison from one trained model per method, this one included, should be read with that specific caveat in mind unless the comparison has itself been replicated.

6 Threats to Validity

Internal validity. This threat is no longer hypothetical for this paper: we tested it directly by retraining MC Dropout three more times (seed 13 again, then genuinely different seeds 17 and 29), all under the identical dropout rate, epoch budget, and manifest used for Table 1. The four Mendeley LBC risk-coverage AUC values – 0.285 (Table 1), 0.503, 0.358, 0.359 – span 0.218, with the two seed-17/29 runs landing in between the two nominally-seed-13 runs rather than clustering near either. This large enough swing erases the reported severe-shift advantage entirely and makes at least one retrained run statistically indistinguishable from the deep ensemble (paired bootstrap 95% CI for the difference $[-0.066, 0.010]$, one-sided $p = 0.929$; Section 4). The deep ensemble's 5 members use one fixed set of seeds, reused unchanged for this significance test; seed-to-seed ensemble variance was not separately measured, so its risk-coverage AUC at each condition (Table 1) is a point estimate from one specific ensemble composition, not an average over repeated ensemble constructions. Table 1's headline Mendeley LBC ranking (MC Dropout 0.285, evidential 0.476, ensemble 0.473) should accordingly be read as the result observed for one specific trained instance of each method, not as a stable per-method property – four independent MC Dropout training runs show this is not a theoretical caveat but an empirically demonstrated failure mode for at least one of the four methods on this exact condition.

External validity. Single backbone (EfficientViT-M0) and two shift conditions (Herlev, Mendeley LBC); the paper's central claim – that uncertainty-method ranking is shift-magnitude-dependent – is demonstrated on this specific backbone and these two shifts, not proven as a general property of uncertainty quantification. Whether the same rank inversion (ensemble best under moderate shift, MC Dropout best under severe shift) would recur with a different backbone, a third shift condition, or a continuous rather than two-point severity scale is untested and is the most direct follow-up this result motivates.

Construct validity. Risk-coverage AUC and selective accuracy at fixed coverage are standard selective-prediction metrics with clear operational meaning (expected error as a function of how much data the system is willing to abstain on), reducing the risk that the rank inversion is an artifact of an unusual metric choice. Evidential deep learning's calibration numbers are reported without the post-hoc correction its raw outputs would need for a fully fair comparison to the other three methods' already-calibrated numbers (Section 4, first paragraph) – this is a known asymmetry in how the four methods are scored, stated explicitly rather than smoothed over.

7 Limitations

Single backbone (EfficientViT-M0); whether this specific inversion (ensemble best under moderate shift, MC Dropout best under severe shift) holds for other architectures is untested. Two shift conditions (Herlev, Mendeley LBC); "moderate" and "severe" are descriptive labels based on the accuracy drop observed, not

independently validated shift-magnitude measurements. Evidential deep learning's calibration numbers are reported without the post-hoc correction its raw outputs would need for a fully fair comparison to the other three methods' already-calibrated numbers. Deep ensemble results depend on the specific 5 seeds used; seed-to-seed ensemble variance was not separately measured. Most importantly, a direct retraining check found MC Dropout's reported severe-shift advantage (Table 1) does not reproduce under an identical seed and recipe (0.503 versus 0.285 risk-coverage AUC on retraining), so the paper's specific severe-shift ranking should not be treated as a stable per-method result – only the higher-level conclusion, that uncertainty-method ranking is shift-dependent and not safely assumed fixed, is expected to generalize beyond the single trained model each method's row in Table 1 represents.

8 Conclusion

Comparing four uncertainty-quantification methods' own cheapest native confidence signals on a backbone with a documented real domain-shift failure shows that method ranking inverts between moderate and severe shift in our original run: deep ensembles win under moderate shift, MC Dropout wins under severe shift, and neither method's advantage transfers to the other condition. A follow-up significance test then complicated this result rather than confirming it: retraining MC Dropout under an identical seed and recipe did not reproduce its reported severe-shift advantage at all, and the retrained run is statistically indistinguishable from the deep ensemble. No method reaches a reliable abstention operating point under the severe shift condition that matters most, in either run. The practical conclusion now has two layers rather than one: a single fixed uncertainty-quantification method, chosen and validated once, cannot be assumed to remain the right choice as shift severity varies; and, more sharply, a single trained instance of that method, evaluated once under one severe-shift condition, cannot be assumed to represent that method's behavior even under an identical seed and recipe. Both need to be tested, not assumed, for any deployed system relying on model confidence to decide when to abstain – and any single-run method comparison, including this paper's own headline numbers, should be read with that caveat attached.

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